

MODERN ENTERPRISE DATA ARCHITECTURE

HEALTHCARE DATA PLATFORM

From Data Chaos to Smart Fraud Detection

SSIS Pipeline

dbt Modeling

N8N & ML Fraud
Detection

cloud pipeline

| What We'll Cover Today

01 The Business Problem

Why healthcare data is broken across providers, insurers, and patients.

02 Solution & Value

Business impact first — speed, accuracy, security, and cost savings.

03 Architecture Overview

End-to-end data flow from raw source to business insight.

04 Technology Stack

ETL → ELT → ML → Alerts, powered by best-in-class tooling.

05 Analytics & Insights

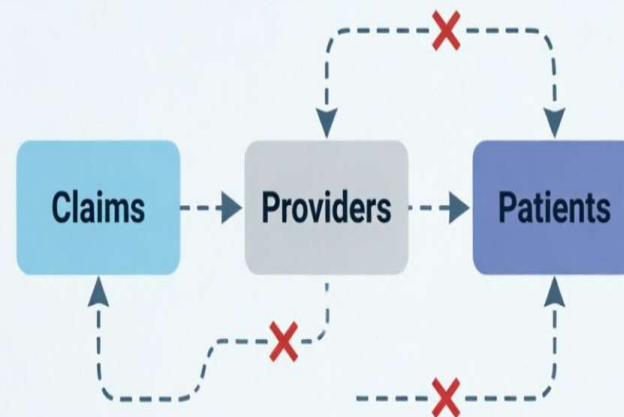
Dashboards, geographic patterns, and fraud detection in Power BI.

Healthcare Data is Broken

THE BUSINESS PROBLEM

- ❌ **Siloed Ecosystem:** Claims data remains heavily scattered across providers, insurers & patients — no integration.
- ❌ **Legacy Formats:** Heavy reliance on disparate CSV files and siloed systems that don't communicate.
- ❌ **Operational Gaps:** Delayed, inaccurate reports leading to wrong business decisions.
- ❌ **Revenue Leakage:** Fraudulent claim activities go undetected, costing companies millions annually.

Fragmented Healthcare Data Ecosystem



| What We Deliver for Your Organization

STRATEGIC CORPORATE BENEFITS



Speed

Process massive healthcare datasets in record time, eliminating engineering bottlenecks.



Accuracy

Clean, standardized data models power trusted, reliable business reports.



Security

Detect fraudulent claims and protect HIPAA-sensitive data before they cost you money.



Cost Savings

Stop fraudulent reimbursements early — protect your bottom line.



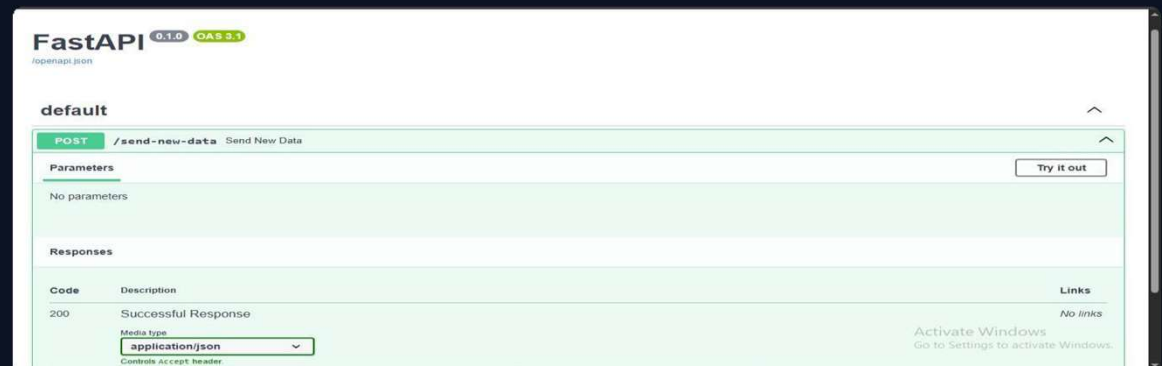
Insights

Real-time Power BI dashboards enable confident, data-driven decisions.

| Extract DATA: API Payloads

FASTAPI INGESTION ENDPOINT

- ✓ **Real-time Extraction:** Payloads are continuously pushed into pipelines using Kafka stream queues.
- ✓ **Standardized API:** Uses a POST /send-new-data endpoint schema.
- ✓ **Fast Buffering:** Directly writes payload streams into the secure local ODS.

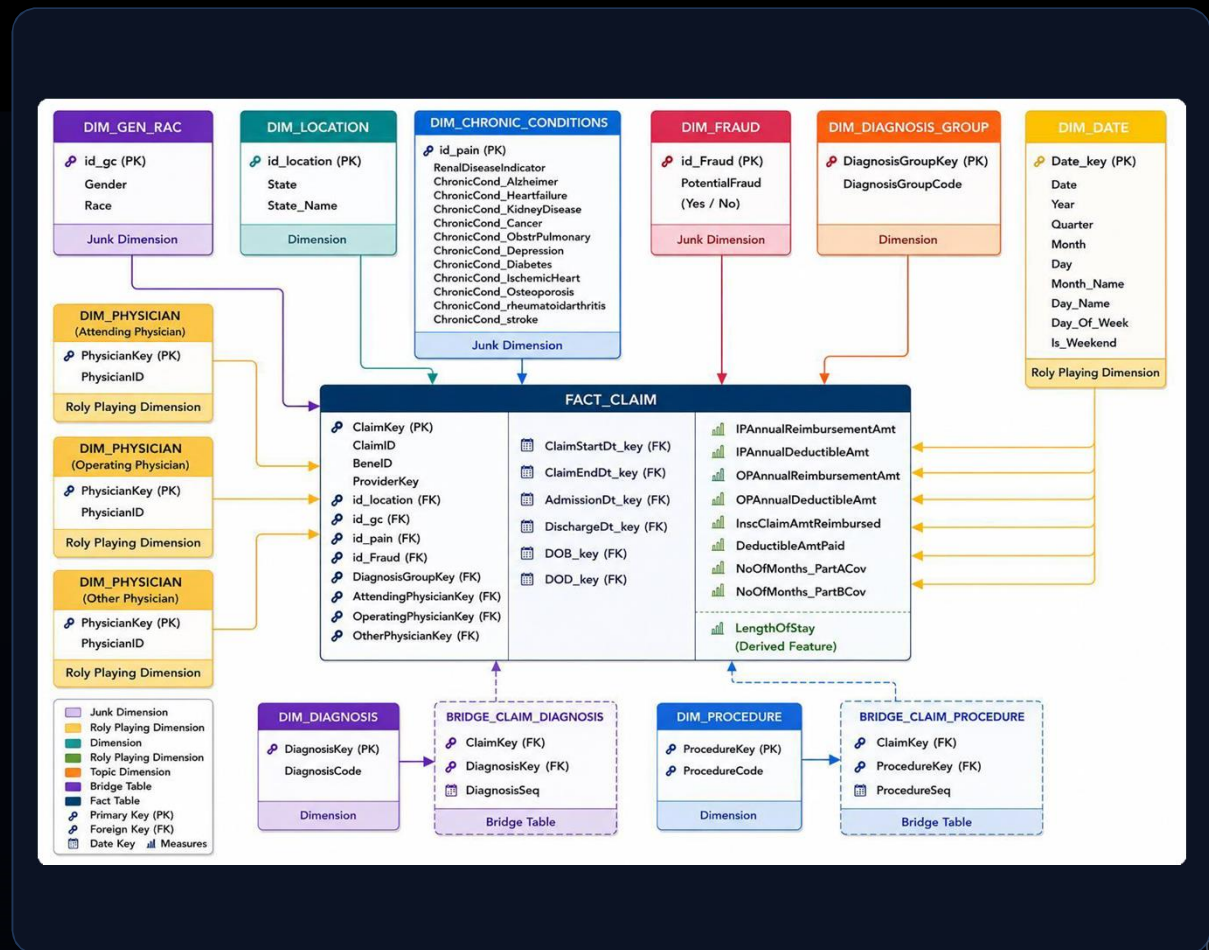


```
curl -X 'POST' '/send-new-data' -H 'accept: application/json'
```

Star Schema — FACT_CLAIM at the Center

FACT_CLAIM AT THE CENTER

- ✓ **Primary Core:** Fact table aggregates reimbursement and deductible metrics.
- ✓ **Dimensions:** Conformed tables handle Physician, Patient, Location, and Date data.
- ✓ **Bridges:** Direct bridge tables manage Claim-to-Diagnosis mapping keys.
- ✓ **Optimization:** Structured strictly for high-speed multi-dimensional analytics.



| Our Solutions

THREE INGESTION & TRANSFORMATION PATHS

SSIS Pipeline

Solution 1

On-premise ETL tool extracting raw feeds directly into a SQL Server Data Warehouse.

CSV → Kafka → Snowflake

Solution 2

Raw CSV source files streamed through a Kafka layer into the Snowflake data warehouse.

Snowflake → Databricks

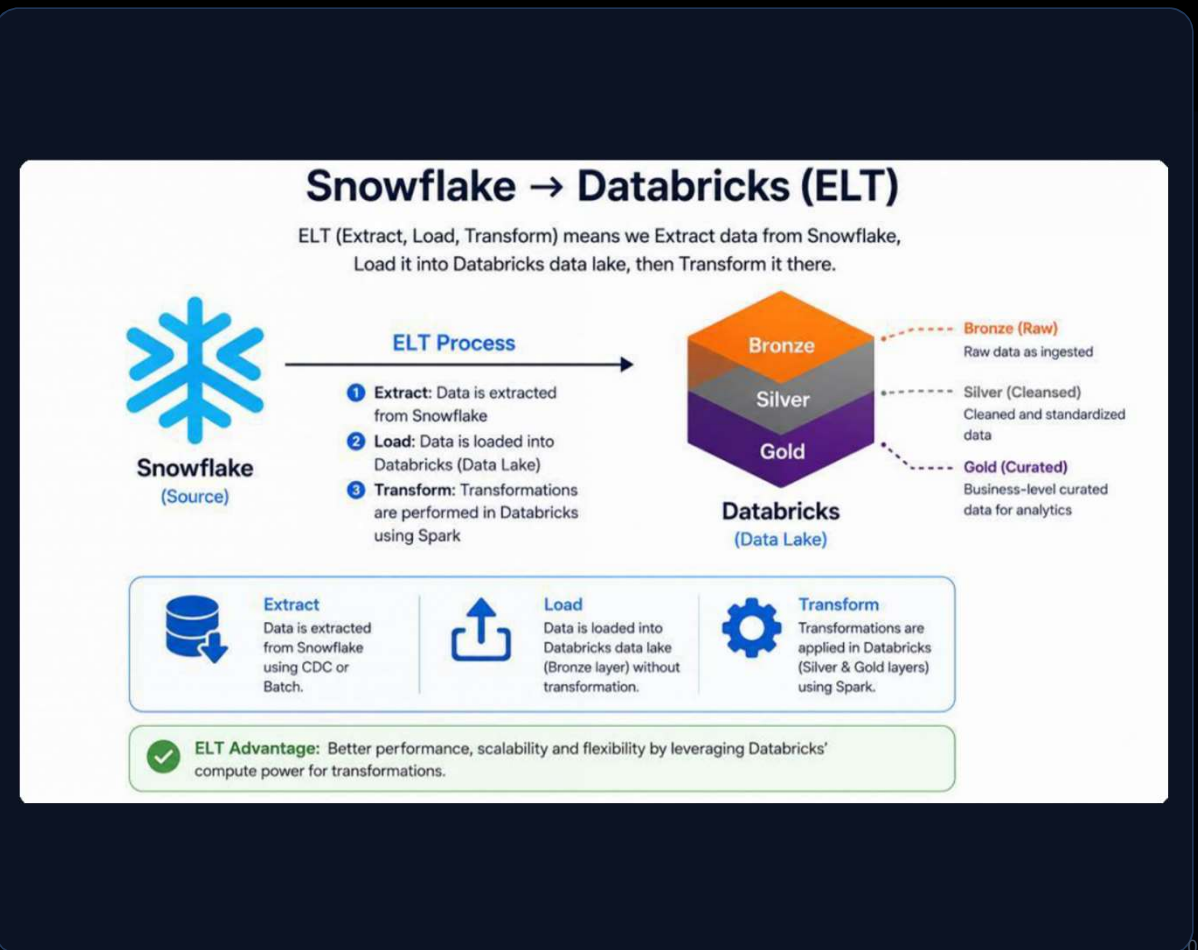
Solution 3

Snowflake as the data source, transformed and analyzed inside the Databricks lakehouse.

Databricks Medallion Lakehouse

BRONZE → SILVER → GOLD ELT PIPELINE

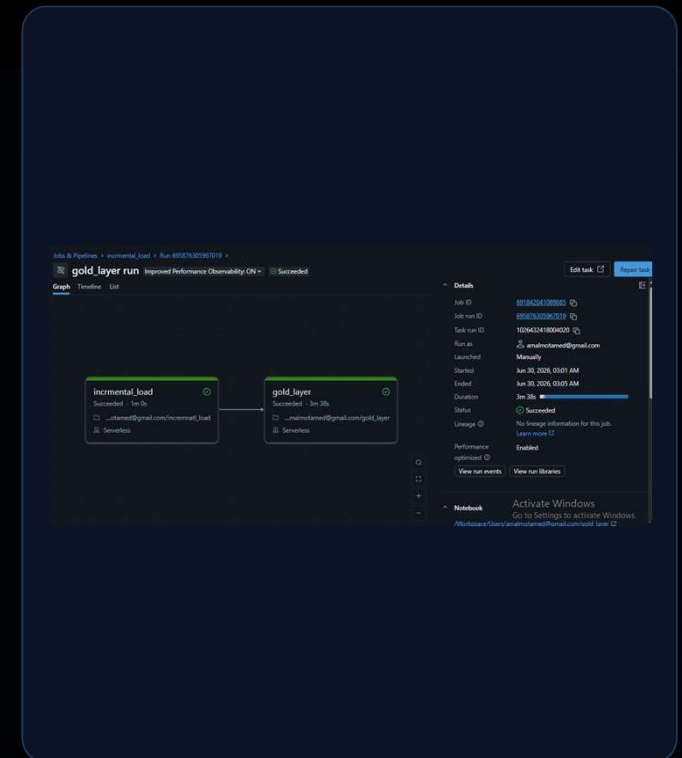
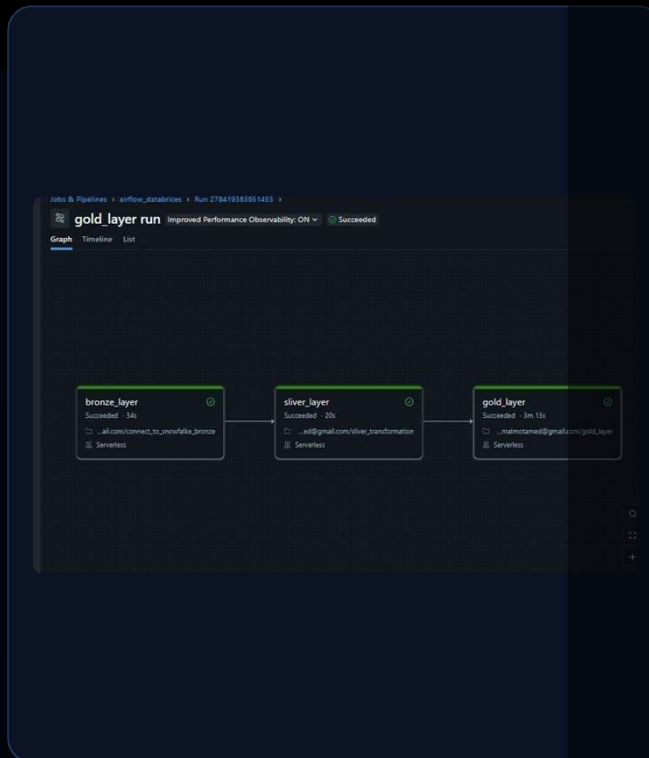
- ✓ **Bronze (Raw):** Loads source files unchanged to preserve historical trace audits.
- ✓ **Silver (Cleansed):** Cleans, de-duplicates, and standardizes records for BI operations.
- ✓ **Gold (Curated):** Highly modeled structures aggregate clean clinical metrics for predictive models.



Databricks Job Executions

PIPELINE OBSERVABILITY

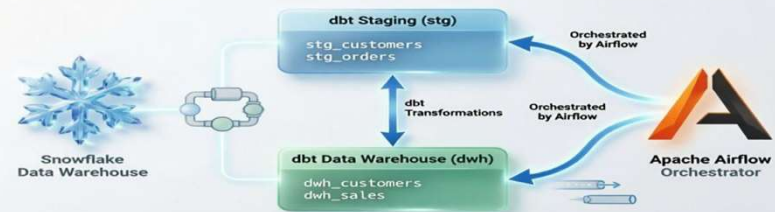
- ✓ **Continuous Tracking:** Full visibility over Medallion pipeline workloads.
- ✓ **Performance Logging:** Active monitoring maps Bronze, Silver, and Gold execution stages.
- ✓ **Succeeded Jobs:** Displays successful runs of incremental loading tasks.



| dbt Models in Snowflake — Staging to DWH

STAGING TO DWH

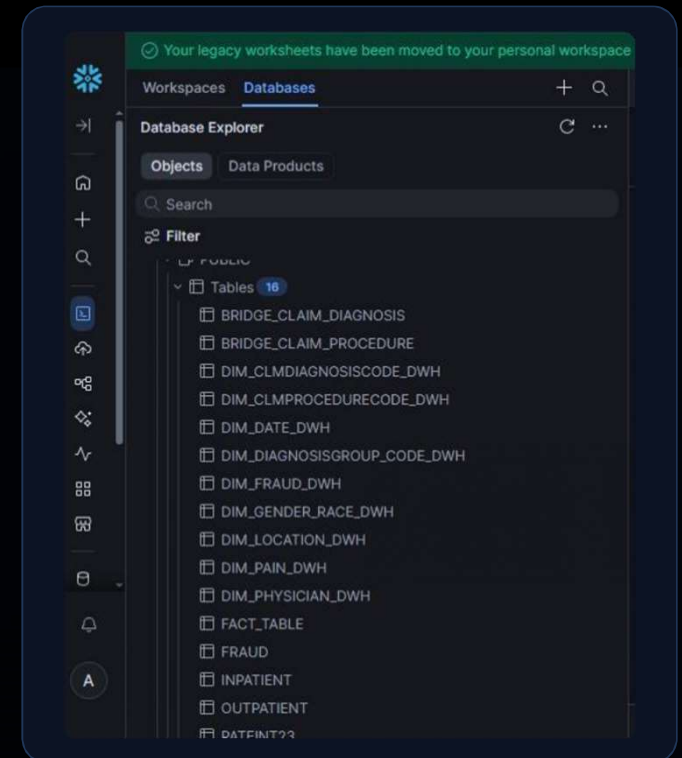
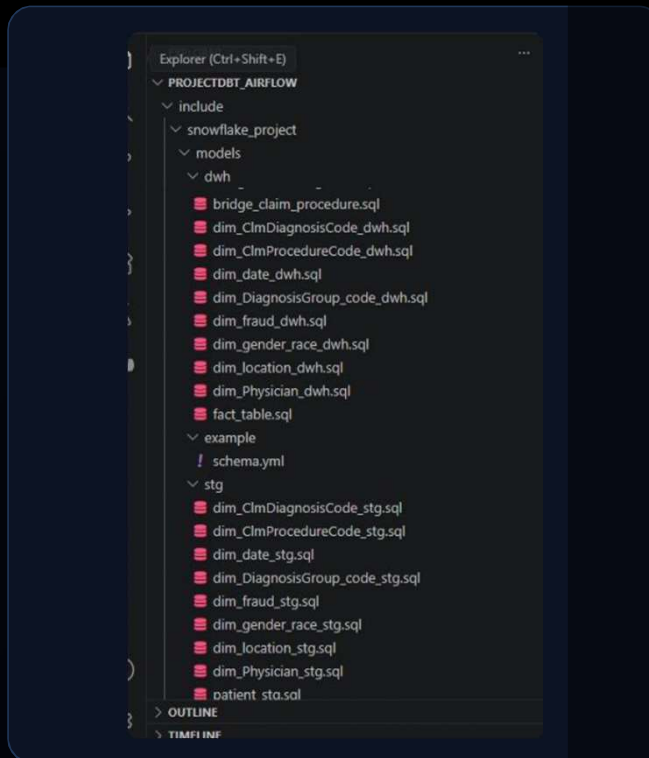
- ✓ **dbt Staging (STG):** Formats transactional structures into clean tables.
- ✓ **dbt Warehouse (DWH):** Structures fully optimized dimensional sets.
- ✓ **Airflow Orchestration:** Airflow orchestrates all dbt runs on schedule.



| dbt + Snowflake

ORCHESTRATION & METADATA CONFIGURATION

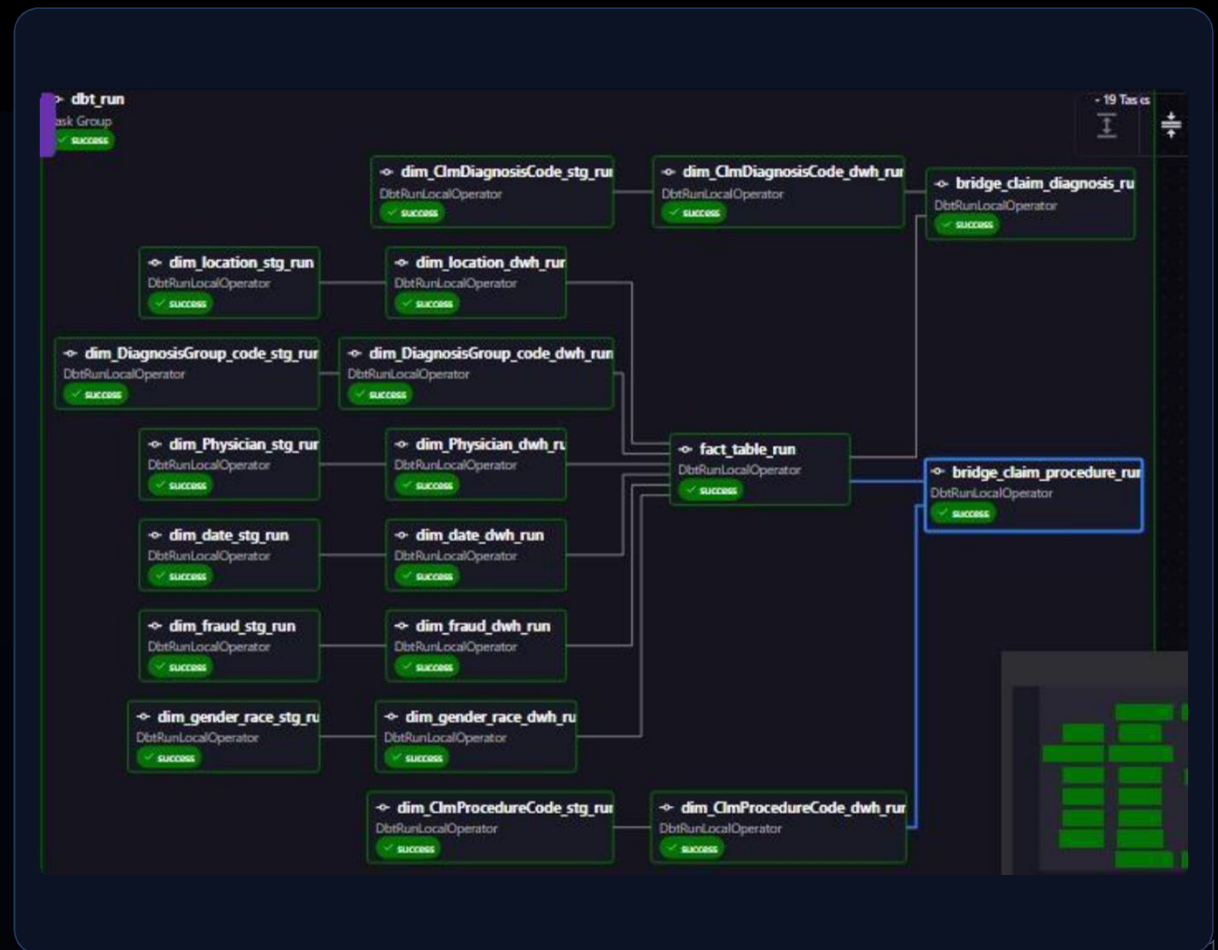
- ✓ **dbt (Data Build Tool):** Manages ODS → STG → DWH transformations inside Snowflake, testing data quality at every step.
- ✓ **Snowflake:** Central warehouse in the Snowflake → dbt → Airflow orchestration chain.



Apache Airflow DAG Orchestration

END-TO-END TASK MANAGEMENT

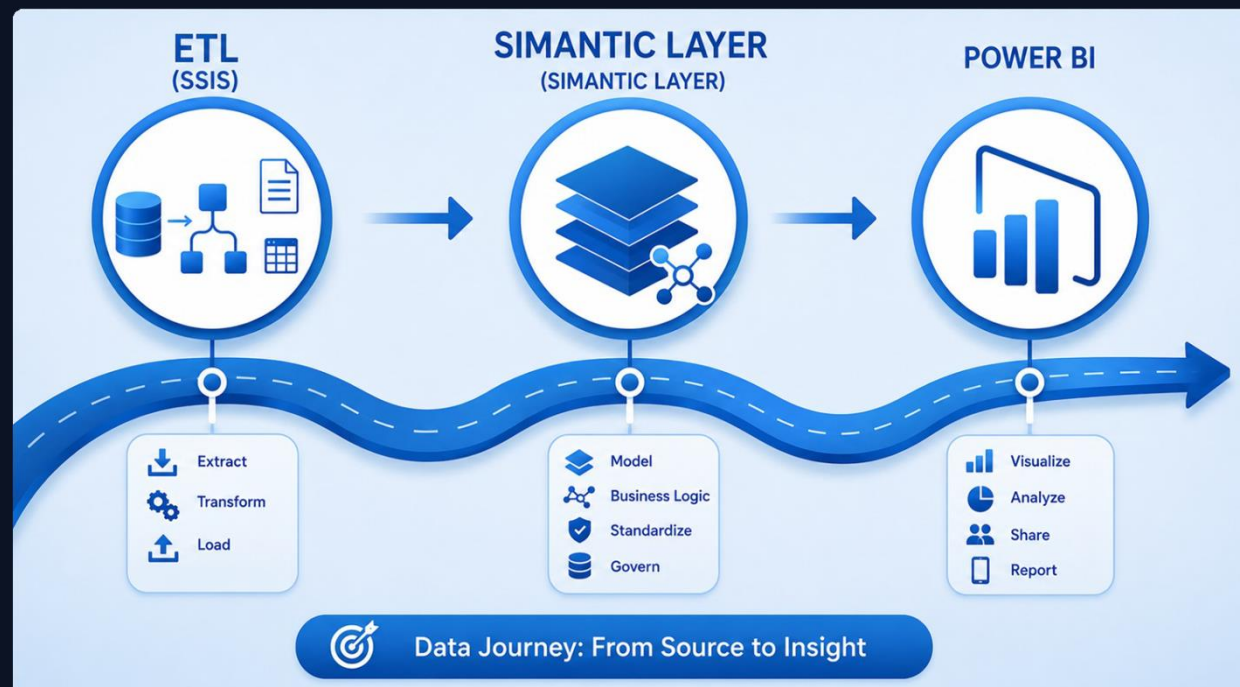
- ✓ **Automated Execution:** Orchestrates the entire pipeline — schedules all tasks automatically.
- ✓ **Task Pipelines:** Manages dependency graphs between staging runs and warehouse loads.
- ✓ **Status Observability:** Monitors every step with full visibility and alerting.



| Data Journey: From Source to Insight

ARCHITECTURE OVERVIEW

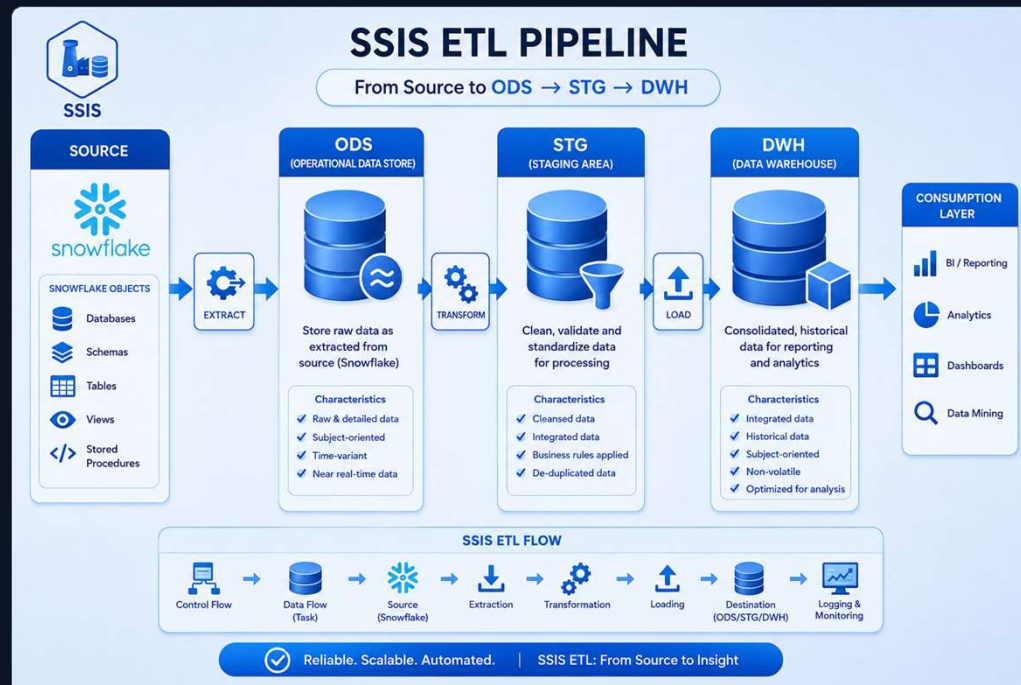
ETL → Semantic Layer → Power BI — the complete path from raw data to business insight



SSIS ETL Pipeline

DATA INGESTION

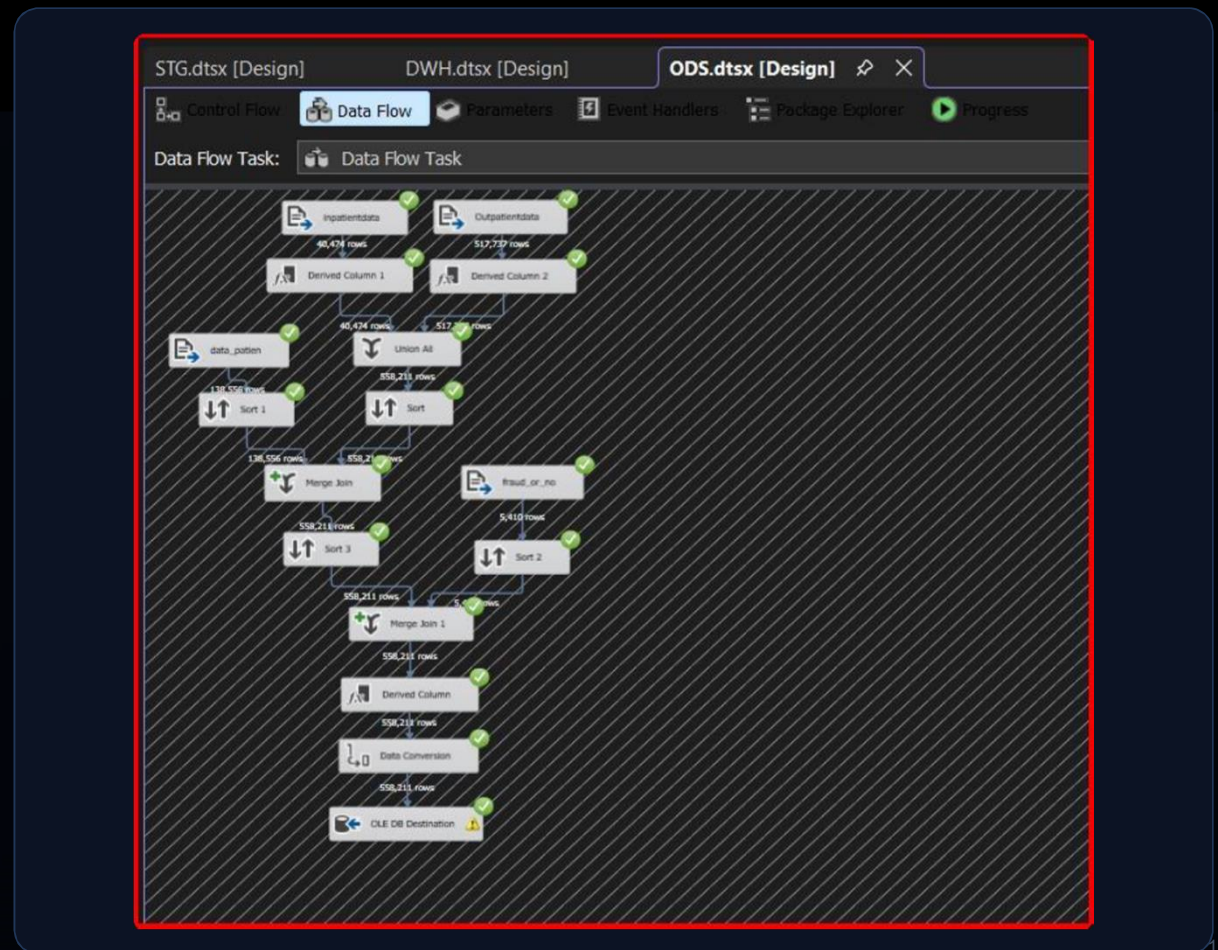
From Snowflake source through ODS → STG → DWH to the consumption layer



ODS Data Flow

SSIS ODS.DTSX PIPELINE

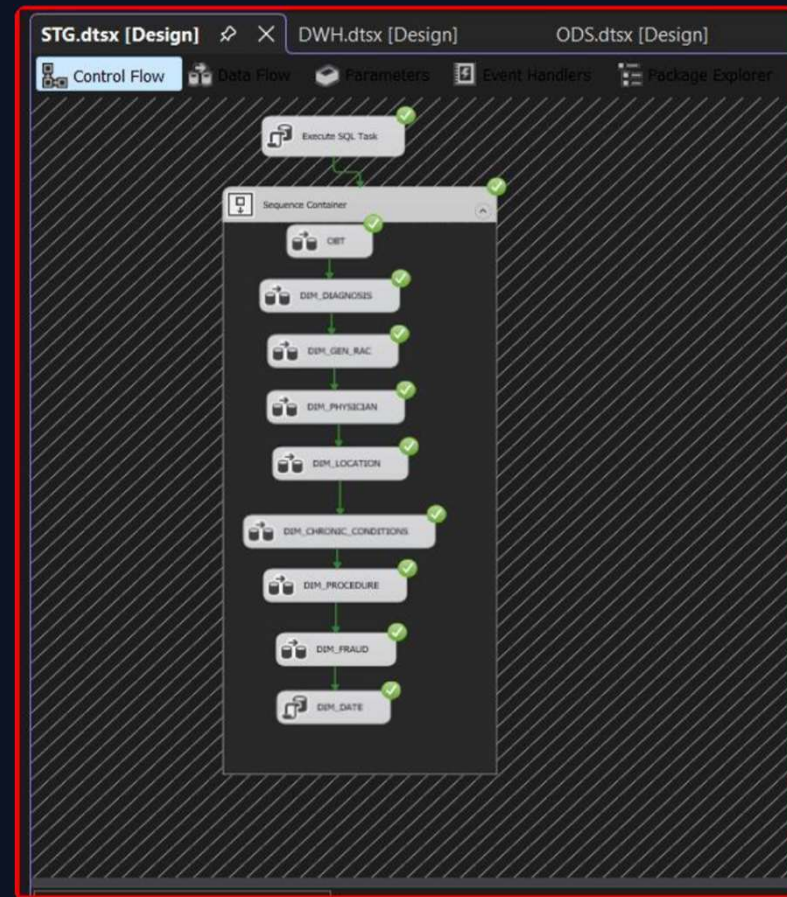
- ✓ **Flat File Extract:** Extracts source datasets directly from provider file locations.
- ✓ **Data Integration:** Integrates inpatient and outpatient records seamlessly.
- ✓ **ODS Checkpoint:** Safely caches raw transactions without database transformations.



Staging Control Flow

SSIS STG.DTSX PACKAGE

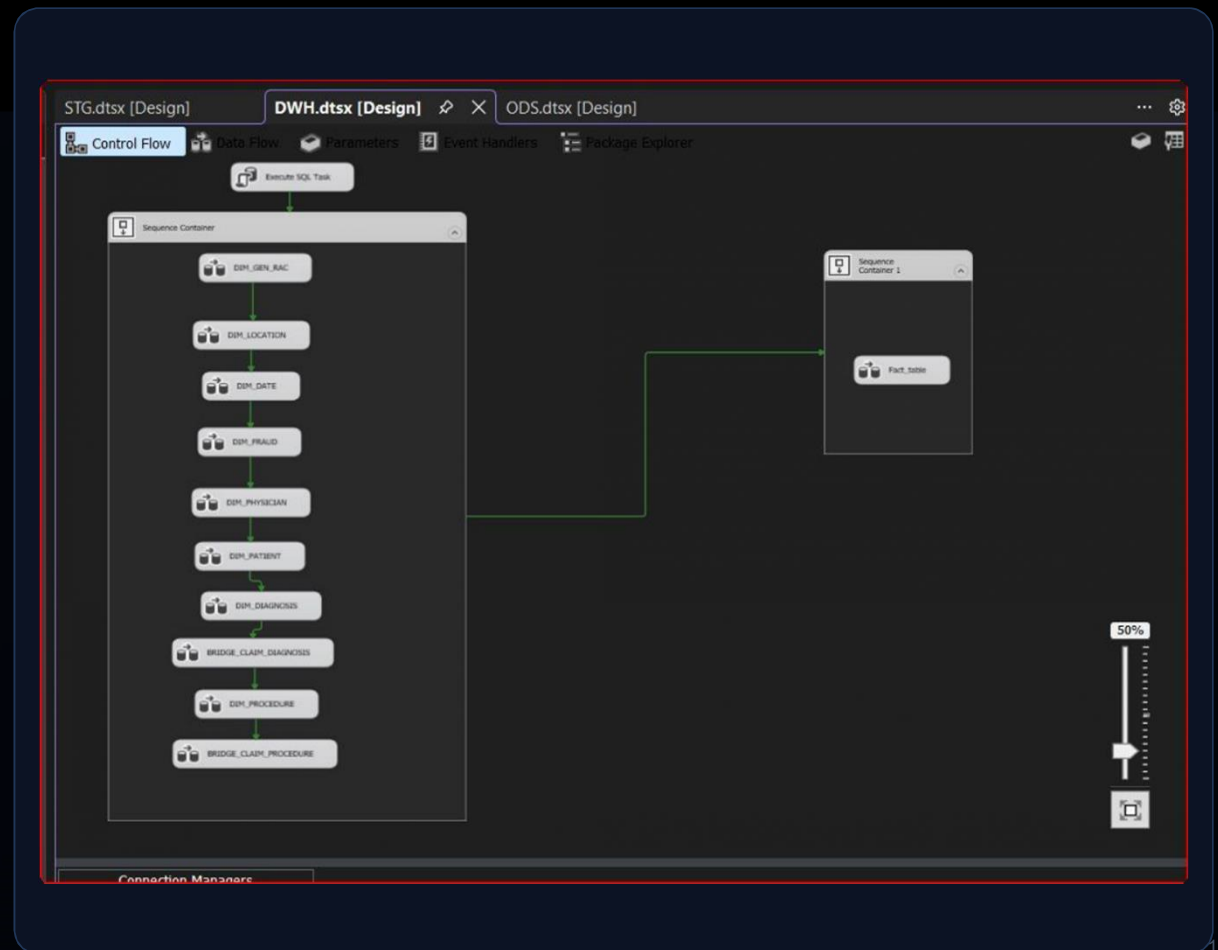
- ✓ **Sequence Container:** Triggers parallel staging loads for conformed dimensions.
- ✓ **SQL Execution:** Drops indices and truncates transient buffers dynamically.
- ✓ **Data Ingest:** Loads clean, validated entries for modeling phases.



DWH Control Flow

SSIS DWH.DTSX PACKAGE

- ✓ **Orchestrated Sequencing:** Executes dimension loads before loading the central fact table.
- ✓ **Data Integrity:** Preserves foreign-key relationships across conformed tables.
- ✓ **Error Handlers:** Automated alerts map job run parameters.



DWH – Fact Table

SSIS DATA FLOW TASK

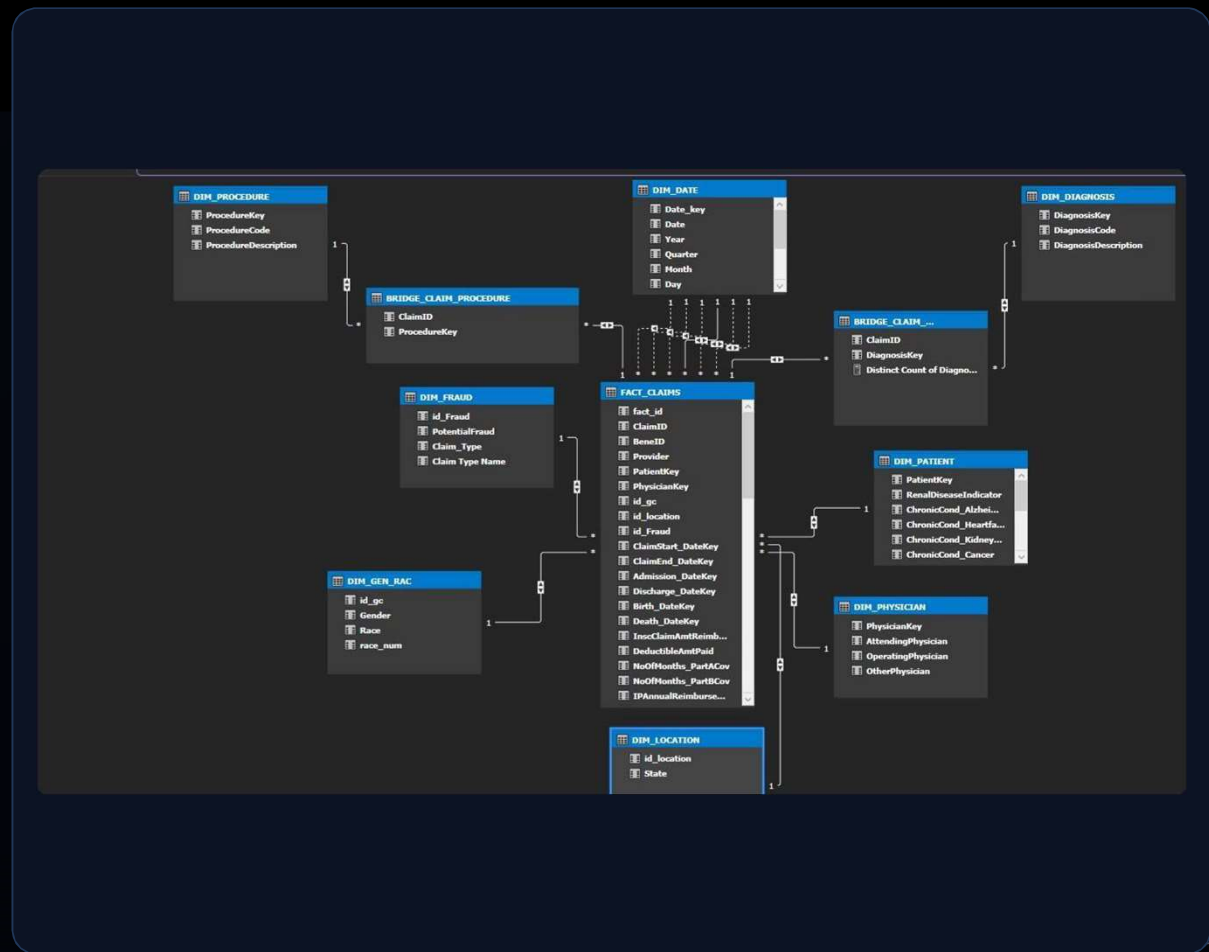
- ✓ **Lookup Pipeline:** Resolves conformed IDs across Location, Date, and Patient dimensions.
- ✓ **Validation Rules:** Mitigates processing bottlenecks across incoming records.
- ✓ **Destination Load:** Loads clean operational records into OLE DB warehouses.



SSAS Tabular Semantic Model

VISUAL BUSINESS MODEL

- ✓ **Unified Access:** Delivers conformed schema relationships for clinical analytics.
- ✓ **Integrated Logic:** Implements central metrics inside conformed columns.
- ✓ **Reporting Ready:** Connects directly to enterprise visualization layers.



Power BI DashBoard

ANALYTICS INTERFACE

- ✓ **Interactive Navigation:** Directly routes users to specific analytics views.
- ✓ **Integrated Dashboards:** Includes Inpatient, Outpatient, and Fraud views.

Healthcare Provider Fraud Detection Analysis

An End-to-End Data Engineering Solution
for Healthcare Fraud Detection

-  Overview
-  Inpatient Analysis
-  Descriptive Analysis
-  Outpatient Analysis
-  Fraud Detection



Overview

SYSTEMIC CLAIM ANALYSIS

38.12%

SYSTEM FRAUD RATE

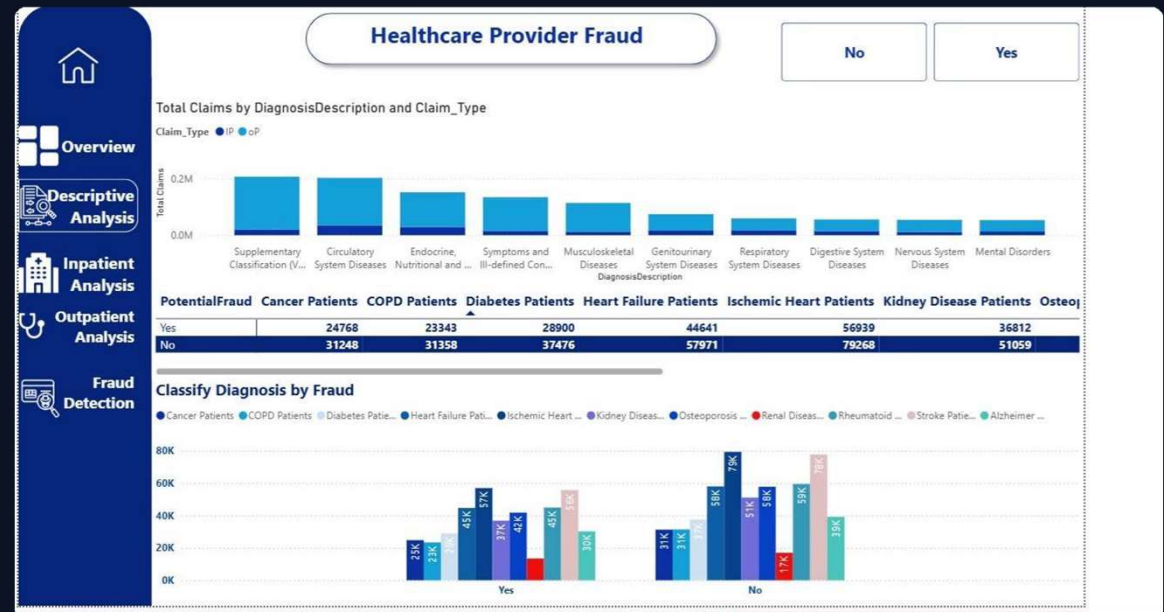
- ✓ **Key KPIs:** Highlights total reimbursed amounts and average claim costs.
- ✓ **Total Reimbursed:** Tracks \$555.00M in overall system claim activities.



Descriptive Analysis

DIAGNOSES PATTERNS

- ✓ **Potential Fraud Flag:** Identifies potential fraud distribution across clinical conditions.
- ✓ **Claims by Diagnosis:** Identifies high claims counts in chronic condition categories.
- ✓ **Detailed Matrix Views:** Maps claims rates to specific medical parameters.



Inpatient Analysis

INPATIENT COST TRENDS

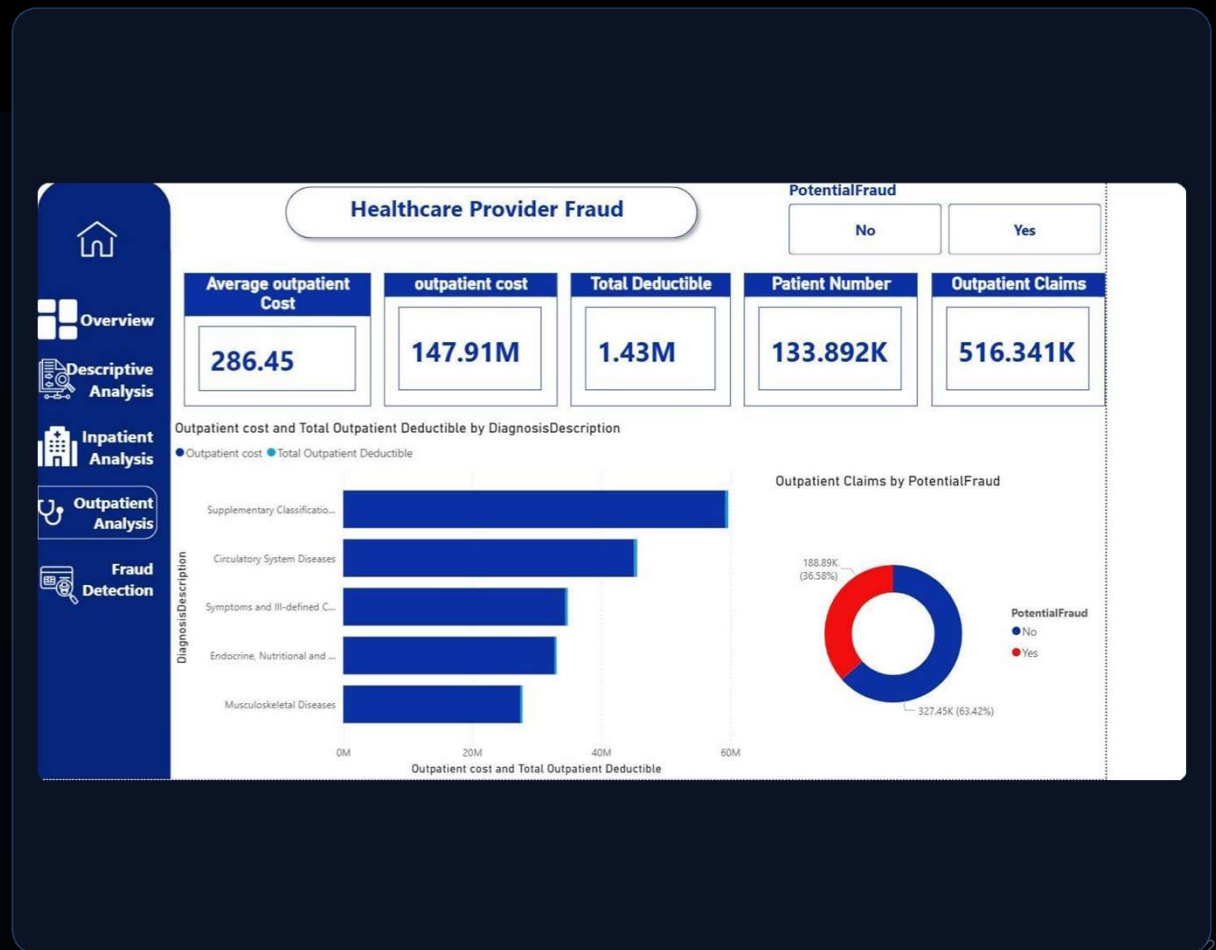
- ✓ **Financial Metrics:** Tracks average inpatient costs alongside total deductibles.
- ✓ **Diagnosis Impact:** Highlights high-cost indicators, led by Circulatory System Diseases.
- ✓ **Outlier Identification:** Isolates outlier claims using advanced visualizations.



| Outpatient Analysis

OUTPATIENT COST TRENDS

- ✓ **Volume Trends:** Evaluates outpatient costs, totaling \$147.91M across clinical providers.
- ✓ **Patient Numbers:** Measures high patient volumes, exceeding 133K outpatient claims.
- ✓ **Diagnosis Categories:** Maps outpatient claims rates against key diagnostic categories.



Fraud Analysis

PROVIDER RISK SCORING

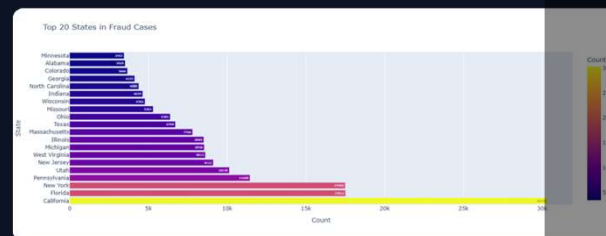
- ✗ **Risk-Scored Providers:** Maps claim patterns to specific flagged providers.
- ✗ **Statistical Insights:** Tracks 506 identified high-risk fraudulent providers.
- ✗ **Financial Impact:** Monitors \$295.01M in flagged fraudulent claims.



Financial Insights — Claims Analysis by State

ANALYTICS

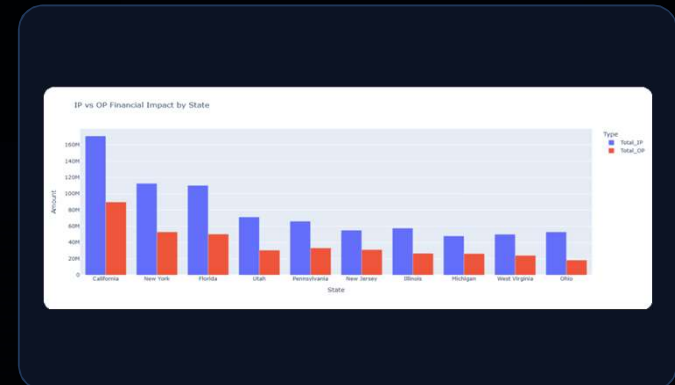
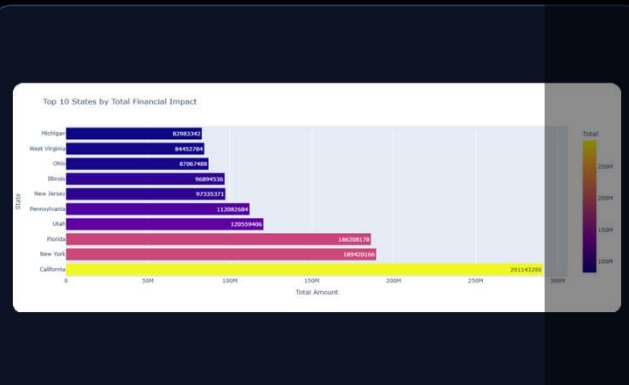
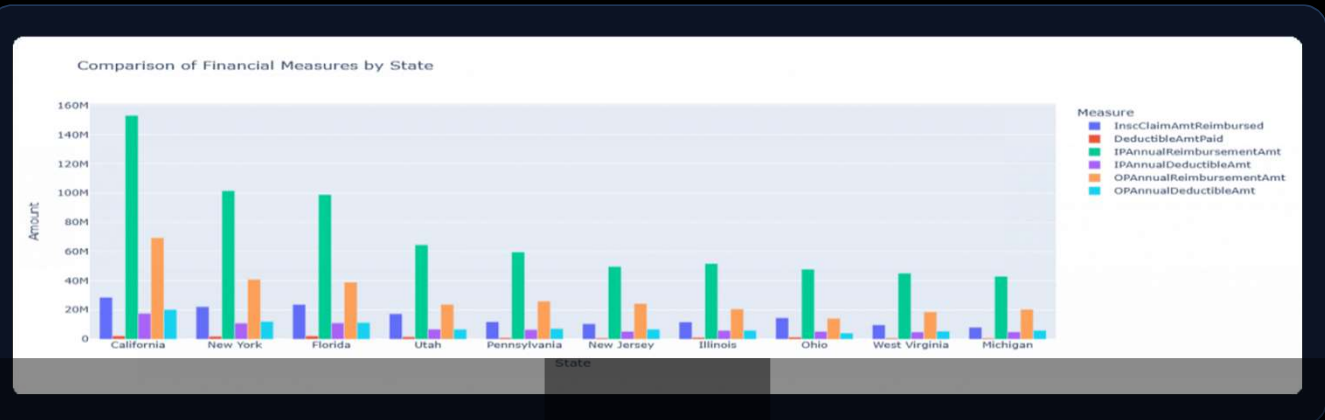
- ✓ **Demographics:** Evaluates gender distribution in identified fraud cases.
- ✓ **Geographics:** Tracks claim distribution and count indexes by state.
- ✓ **Race Distribution:** Breaks down fraud cases by patient race.



Fraud Pattern Analysis — Geographic & Demographic

FRAUD ANALYTICS

- ✓ **Measures Contrast:** Evaluates annual reimbursement rates across state boundaries.
- ✓ **State Metrics:** Highlights California and New York as top financial impact states.



| Feature Engineering — RFE

RECURSIVE FEATURE ELIMINATION

RANK	FEATURE NAME	TYPE	SCORE
1	Provider	Categorical	0.128
2	ChronicCond_IschemicHeart	Binary	0.104
3	ChronicCond_Diabetes	Binary	0.098
4	NoOfMonths_PartACov	Numeric	0.085
5	ChronicCond_Heartfailure	Binary	0.076
6	Gender	Categorical	0.069
7	NoOfMonths_PartBCov	Numeric	0.061
8	State	Categorical	0.058

RFE Approach

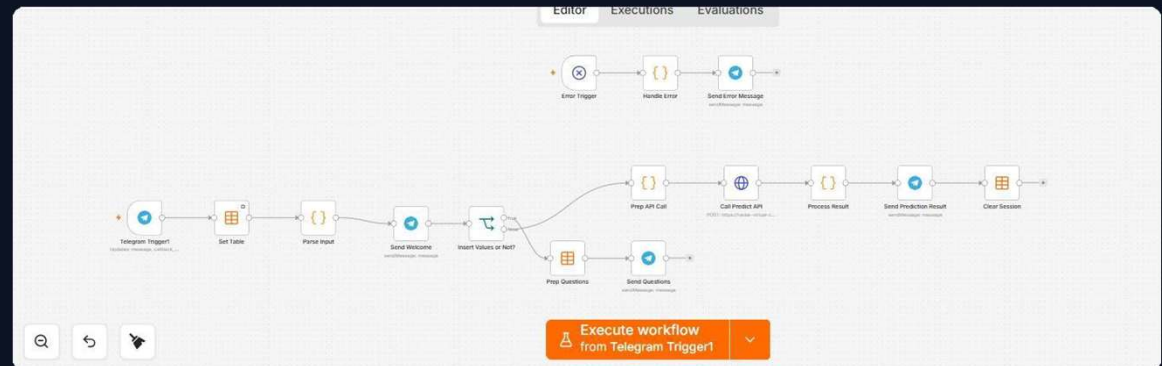
Recursive Feature Elimination (RFE) selects the most relevant features to improve model performance while preventing over-fitting.

20 features selected from 20 total candidates.

Real-Time Fraud Detection Pipeline

REAL-TIME ANOMALY ROUTING

- ✓ **Gold Layer (Databricks):** Supplies clean patient and claims records to predictive models.
- ✓ **ML Model (Predict API):** Evaluates incoming records to generate real-time risk scores.
- ✓ **n8n Workflow → Telegram:** Orchestrates tasks to route flagged anomalies straight to a Telegram alert.



| Best-in-Class Tools Working Together

TECHNOLOGY STACK

SSIS

ETL Orchestration — extracts raw CSV files and structures local Operational Data Stores.

SQL Server

ODS / Staging — hosts the ODS and Staging layers for raw and cleansed clinical data.

Snowflake

Data Warehouse — the centralized cloud data warehouse and analytical repository.

dbt

Data Transformation — standardizes staging transformations and models the fact table.

Airflow

Pipeline Orchestration — orchestrates end-to-end schedules and pipeline tasks.

Databricks

Lakehouse / Spark — powers the Bronze → Silver → Gold ELT pipeline.

FastAPI

Data Ingestion API — provides the real-time data ingestion endpoint.

Power BI

Dashboards & Reports — renders conformed SSAS tabular models into dashboards.

n8n + ML

Fraud Alerts — leverages predictive APIs to generate instant fraud alerts.



Ready to Transform Your Healthcare Data?

Deploy a modern, scalable, and secure clinical data pipeline to protect revenue lines and drive better organizational decisions every day.

- ✓ Modern, scalable data platform built for healthcare
- ✓ Real-time fraud detection that protects your revenue
- ✓ Clean, reliable data powering better decisions every day

Let's Build the Future Together

THANK YOU